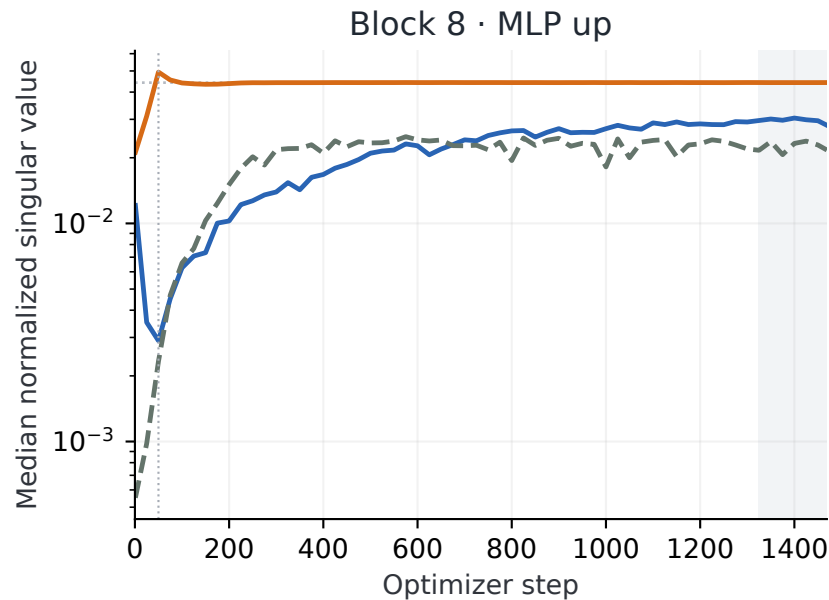
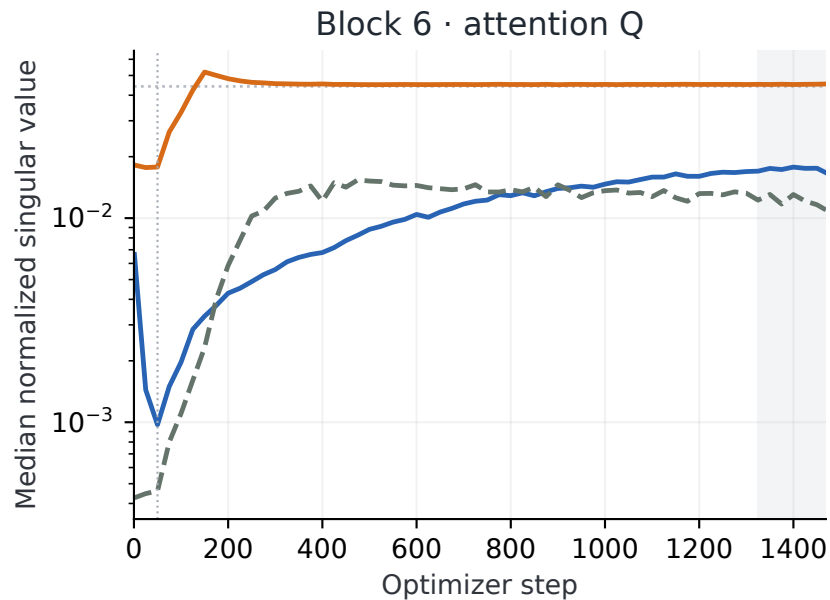
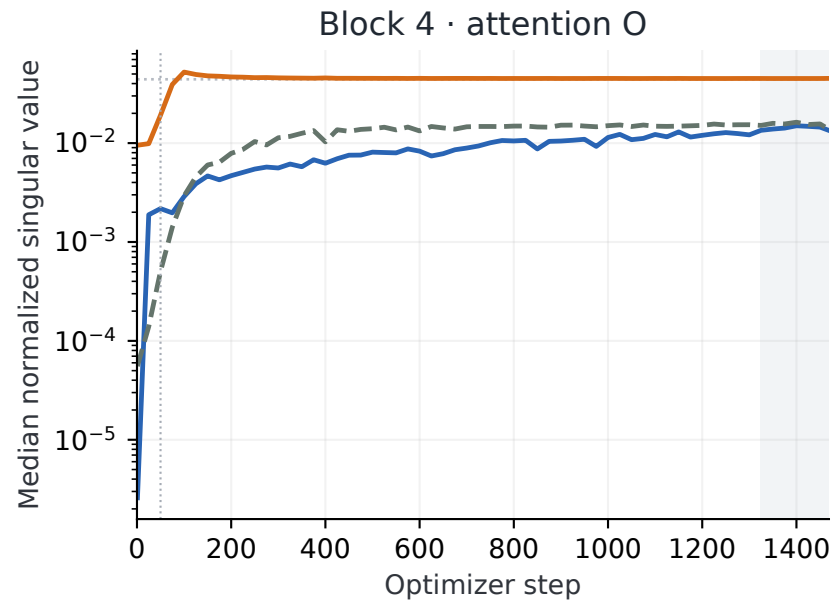


8 layers · spectral evolution at three fixed locations

AdamW adaptive update Muon post-NS update Muon pre-NS momentum



Each matrix is divided by its own Frobenius norm. Dashed green is momentum, before NS; solid curves are updates. Dotted horizontal line: equal singular values, $1/\sqrt{512} \approx 0.0442$. LR and weight decay are excluded.